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GROUP 13

COMP8325 Applications of Artificial Intelligence for Cyber Security

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Task 1: Model Training

1. Introduction

This report presents the training and evaluation of two machine learning classifiers, Random Forest and XGBoost, for multi-class malware classification using the BODMAS dataset. The task focuses on classifying Windows Portable Executable files into benign and malware categories using static PE feature vectors. The dataset contains a large number of numerical features and highly imbalanced class labels, making preprocessing and class-level evaluation important. The models were trained on a stratified training set and evaluated on a held-out test set. Performance was assessed using overall accuracy, confusion matrices, True Positive Rate, and False Positive Rate.

2. Dataset Description

The BODMAS dataset is a labelled malware analysis dataset containing Windows Portable Executable (PE) files represented as numerical static-analysis feature vectors. Each sample is described using 2,381 features extracted with the LIEF library, allowing machine learning models to classify files without executing them. The dataset contains three main files: `bodmas.npz`, which stores the feature matrix and binary labels; `bodmas_metadata.csv`, which provides SHA-256 hashes and metadata aligned with the feature rows; and `bodmas_malware_category.csv`, which provides malware category labels for malicious samples. Since benign samples are not included in the malware category file, they are labelled separately using the binary label. The final dataset contains 134,435 samples across 15 classes, including one benign class and 14 malware categories. The dataset is highly imbalanced, with large classes such as benign, trojan, and worm, and very small classes such as rootkit and trojan-gamethief.

3. Data Preprocessing

3.1 Label Preparation

The preprocessing stage first combined the BODMAS feature vectors with the correct category labels. The feature matrix and binary labels were loaded from `bodmas.npz`, while sample metadata was taken from `bodmas_metadata.csv`. The actual malware category labels were obtained from `bodmas_malware_category.csv`, because this file provides the target malware categories for the multi-class classification task. The metadata and malware category files were merged using the SHA-256 hash as the common identifier, ensuring each malware sample was matched with its correct category.

Since `bodmas_malware_category.csv` only contains malware samples, benign samples did not have category labels in that file. To create a complete 15-class dataset, all rows with a binary label 0 were assigned the category label benign. Malware samples with binary label 1 were assigned their corresponding category from the malware category file.

3.2 Feature Cleaning and Label Encoding

The BODMAS feature matrix contains 2,381 numerical static-analysis features extracted from Windows Portable Executable files using the LIEF library. Since machine learning models require valid numerical input, the feature matrix was checked for missing and invalid values before training. Any NaN, positive infinity, or negative infinity values were replaced with 0.0 using `np.nan_to_num()`. This prevented runtime errors and allowed both classifiers to use the complete feature matrix. Low-variance feature selection was applied using `VarianceThreshold`, reducing the feature set from 2,381 to 1,501 features. The fitted selector was saved and reused for the holdout dataset to ensure consistent preprocessing.

The category labels were originally stored as text values such as benign, trojan, worm, backdoor, and ransomware. Since Random Forest and XGBoost require numerical target labels, these class names were converted into integer labels using `LabelEncoder`. The encoder mapping was retained so model predictions could later be converted back into readable malware category names during evaluation.

3.3 Train-Test Split

The prepared dataset was divided into training and testing sets using an 80:20 split. A stratified split was used to preserve the class proportions in both sets, which was important because the BODMAS dataset is highly imbalanced. Large classes, such as benign, trojan, and worm have many samples, while minority classes, such as rootkit, trojan-gamethief, and exploit have very few samples. The split was performed with a fixed random state of 42 to ensure reproducibility. The training set was used to train the Random Forest and XGBoost classifiers, while the test set was kept separate and used only for final evaluation.

4. Models and Training Methodology

4.1 Model 1: Random Forest

Random Forest is an ensemble of decision trees that naturally handles multi-class classification, requires no feature scaling, and is resilient to noisy and correlated features. It also provides feature importance rankings, which can support interpretability. This makes it suitable for the highly imbalanced BODMAS dataset, where large classes such as benign, trojan, and worm dominate, while minority classes such as rootkit and trojan-gamethief contain very few samples.

4.2 Model 2: XGBoost

XGBoost was chosen as the second classifier because it performs strongly on structured tabular data. Unlike Random Forest, which builds trees independently, XGBoost uses gradient boosting, where each new tree attempts to correct the errors made by previous trees. It also includes L1/L2 regularisation to reduce overfitting, making it suitable for the BODMAS dataset, which has a large feature space and class imbalance.

5. Evaluation

The trained Random Forest and XGBoost models were evaluated using the held-out 20% test set, which was not used during model training. Since the BODMAS dataset is highly imbalanced, overall accuracy alone was not sufficient to judge model performance. Therefore, each model was evaluated using overall accuracy, confusion matrix, per-class True Positive Rate (TPR), and per-class False Positive Rate (FPR).

Overall accuracy shows the proportion of correctly classified samples across all 15 classes. The confusion matrix was used to identify which malware categories were correctly classified and which categories were commonly confused. TPR was calculated for each class to measure how many samples from that category were correctly detected. FPR was calculated to measure how often samples from other classes were incorrectly predicted as that category.

This evaluation approach is suitable because the dataset contains large classes such as benign, trojan, and worm, while classes such as rootkit, trojan-gamethief, and exploit have very few samples. Per-class TPR and FPR, therefore, provide a clearer understanding of model behaviour across both majority and minority classes.

6. Results

The overall test accuracy results are shown in Table 5. Random Forest achieved the highest accuracy with 0.9733, slightly outperforming XGBoost, which achieved 0.9703. Random Forest achieved very high training accuracy, but the test accuracy remained strong, suggesting some overfitting while still generalising well. This indicates that both models performed well on the held-out test set, but Random Forest provided the stronger overall classification performance for this dataset.

The detailed per-class TPR and FPR results for both models are shown in Tables 6 and 7.

Task 2: Holdout Dataset

1. Overview

The holdout dataset was used to evaluate how well the two trained classifiers generalised to unseen samples. This stage was completed after Task 1 and used the already trained Random Forest and XGBoost models. No retraining was performed after the holdout dataset was released, so Task 2 measured model generalisation rather than adaptation to the new data.

The holdout dataset contained 34,435 samples, including 18,363 benign samples and 16,072 malware samples. The main categories present were benign, trojan, worm, backdoor, informationstealer, dropper, ransomware, downloader, p2p-worm, pua, trojan-gamethief, and cryptominer. Exploit, rootkit, and virus had no holdout samples, so their TPR values cannot be meaningfully interpreted.

2. Preprocessing

The same preprocessing pipeline from Task 1 was applied to the holdout dataset. The holdout feature matrix was loaded from `bodmas_holdout.npz`, and metadata was loaded from

bodmas_metadata_holdout.csv. Malware category labels were prepared by matching SHA-256 hashes with the holdout malware category file. Since the malware category file does not include benign samples, benign rows were assigned the class label benign using the binary label.

Invalid numerical values, including missing values and infinities, were replaced with 0.0. The saved VarianceThreshold selector from Task 1 was applied to reduce the holdout features from 2,381 to the same 1,501 selected features used by the trained models. The saved LabelEncoder was also reused, so the holdout labels followed the same 15-class mapping used during training and testing.

3. Evaluation Metrics

The holdout evaluation used overall accuracy, True Positive Rate (TPR), and False Positive Rate (FPR). Accuracy measured the proportion of all holdout samples correctly classified, but because the dataset was imbalanced, it was not enough on its own to judge model performance.

TPR measured how well each model correctly identified samples from a specific class, while FPR measured how often samples from other classes were incorrectly predicted as that class. For this multi-class problem, TPR and FPR were calculated using a one-vs-rest approach for each category, including benign.

4. Evaluation Results

The overall holdout accuracy results are shown in Appendix 2, Table 1. Random Forest achieved the best overall result with a holdout accuracy of 0.9950, while XGBoost achieved 0.9781. Both models performed strongly on unseen data, but Random Forest produced the higher overall accuracy.

The per-class TPR and FPR results are shown in Appendix 2, Tables 2 and 3. Random Forest achieved strong TPR values for the major categories, including 1.0000 for benign, 0.9880 for trojan, 0.9899 for worm, and 0.9959 for backdoor. XGBoost also performed well on the major classes, with TPR values of 0.9996 for benign, 0.9259 for trojan, 0.9868 for worm, and 0.9745 for backdoor.

5. Evaluation Performance

Random Forest showed stronger and more consistent performance across the holdout dataset. It also performed better on several smaller categories, with TPR values of 0.9455 for downloader, 0.9556 for dropper, 0.9619 for ransomware, and 0.8462 for p2p-worm.

XGBoost remained a strong classifier overall, but it was weaker on some minority classes. Its TPR values were 0.6364 for downloader, 0.7185 for dropper, 0.8190 for ransomware, and 0.1538 for p2p-worm. This shows that high overall accuracy can hide weaker detection of smaller malware categories. Both models had generally low FPR values, although XGBoost had a higher worm FPR of 0.0142 compared with 0.0021 for Random Forest.

6. Overall Comparative Analysis

The Task 1 and Task 2 accuracy comparison is shown in Appendix 2, Table 4. Random Forest improved from 0.9733 on the original test set to 0.9950 on the holdout dataset, while XGBoost improved from 0.9703 to 0.9781. Since the models were not retrained, this improvement most likely reflects differences in the holdout dataset distribution rather than model improvement.

Overall, Random Forest was the better-performing classifier in Task 2. It achieved higher accuracy, stronger TPR values across most categories, and consistently low FPR values. XGBoost also generalised well, but its weaker detection of smaller categories shows why per-class evaluation is important. Therefore, the holdout evaluation supports Random Forest as the more reliable model, while also showing that class imbalance remains an important limitation.

Appendix 1: Tables for Task 1

Category	Samples	Category	Samples
trojan	29972	virus	192
Worm	16697	pua	29
backdoor	7331	cryptominer	20
downloader	1031	p2p-worm	16
ransomware	821	exploit	12
dropper	715	trojan-gamethief	6
informationstealer	448	rootkit	3
benign	77,142		

Table 1: BODMAS malware category distribution

Split	Samples	Proportion
Training set	107,548	80%
Test set	26,887	20%
Total dataset	134,435	100%

Table 2: Train-Test Split Summary

Parameter	Value	Justification
n_estimators	[50, 100, 200]	More trees improve accuracy. However, it increases training time.
max_depth	[None, 10, 20]	It controls overfitting

Parameter	Value	Justification
max_features	['sqrt', 'log2']	It limits the features per split to reduce correlation between trees.
n_iter	10	This balances search thoroughness with training time
cv	3-fold stratified	It preserves class proportions across folds, which is important for the imbalanced BODMAS dataset.
random_state	42	It controls the shuffling applied to the data before splitting.

Table 3: Random Forest Hyperparameters

Parameter	Value	Justification
n_estimators	[50, 100]	It controls the number of boosting rounds.
max_depth	[3,4]	Shallow trees ask fewer questions, preventing the model from memorising rare cases rather than general patterns.
learning_rate	[0.1, 0.2]	Higher rate converges faster, and balanced with n_estimators
tree_method	hist	It's a histogram-based splitting for faster training on a larger dataset.
n_iter	5	Reduced to limit training time.
cv	2-fold stratified	Faster cross-validation while maintaining class balance
random_state	42	It controls the shuffling applied to the data before splitting.

Table 4: XGBoost Hyperparameters

Model	Overall Accuracy
Random Forest	0.9733
XGBoost	0.9703

Table 5: Overall test set accuracy

Category	Support	TPR	FPR
Benign	15428	0.9998	0.0079
Trojan	5995	0.9328	0.0121
Worm	3339	0.9518	0.0091
Backdoor	1466	0.9707	0.0033
Downloader	206	0.8641	0.0006
Ransomware	164	0.8902	0.0002
Dropper	143	0.7692	0.0016
Info Stealer	90	0.7889	0.0003
Virus	39	0.9744	0.0000
PUA	6	0.1667	0.0000
Cryptominer	4	1.0000	0.0000
P2P-Worm	3	0.0000	0.0000
Exploit	2	1.0000	0.0000
Rootkit	1	0.0000	0.0000
Trojan-Gamethief	1	1.0000	0.0000

Table 6: Random Forest per-class TPR and FPR

Category	Support	TPR	FPR
Benign	15428	0.9964	0.0069
Trojan	5995	0.9271	0.0145
Worm	3339	0.9680	0.0116
Backdoor	1466	0.9686	0.0037
Downloader	206	0.7136	0.0007

Category	Support	TPR	FPR
Ransomware	164	0.8780	0.0003
Dropper	143	0.6853	0.0005
Info Stealer	90	0.7778	0.0003
Virus	39	1.0000	0.0000
PUA	6	0.3333	0.0000
Cryptominer	4	1.0000	0.0000
P2P-Worm	3	0.0000	0.0000
Exploit	2	1.0000	0.0000
Rootkit	1	0.0000	0.0000
Trojan-Gamethief	1	1.0000	0.0000

Table 7: XGBoost per-class TPR and FPR

Appendix 2: Tables for Task 2

Model	Overall Accuracy
Random Forest	0.9950
XGBoost	0.9781

Table 1: Overall Holdout Accuracy

Category	Support	TPR	FPR
Benign	18363	1.0000	0.0008
Trojan	6724	0.9880	0.0026
Worm	6145	0.9899	0.0021
Backdoor	2666	0.9959	0.0003
Downloader	55	0.9455	0.0001

Category	Support	TPR	FPR
Ransomware	105	0.9619	0.0000
Dropper	135	0.9556	0.0003
Info Stealer	217	0.9908	0.0001
Virus	0	0.0000	0.0000
PUA	9	0.7778	0.0000
Cryptominer	1	1.0000	0.0000
P2P-Worm	13	0.8462	0.0000
Exploit	0	0.0000	0.0000
Rootkit	0	0.0000	0.0000
Trojan-Gamethief	2	1.0000	0.0000

Table 2: Random Forest Holdout Per-Class TPR and FPR

Category	Support	TPR	FPR
Benign	18363	0.9996	0.0042
Trojan	6724	0.9259	0.0069
Worm	6145	0.9868	0.0142
Backdoor	2666	0.9745	0.0023
Downloader	55	0.6364	0.0001
Ransomware	105	0.8190	0.0001
Dropper	135	0.7185	0.0002
Info Stealer	217	0.9539	0.0001
Virus	0	0.0000	0.0000

Category	Support	TPR	FPR
PUA	9	0.8889	0.0000
Cryptominer	1	1.0000	0.0000
P2P-Worm	13	0.1538	0.0000
Exploit	0	0.0000	0.0000
Rootkit	0	0.0000	0.0000
Trojan-Gamethief	2	1.0000	0.0000

Table 3: XGBoost Holdout Per-Class TPR and FPR

Model	Task 1 Test Accuracy	Task 2 Holdout Accuracy	Difference
Random Forest	0.9733	0.9950	+0.0217
XGBoost	0.9703	0.9781	+0.0078

Table 4: Task 1 vs Task 2 Accuracy Comparison